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OPEN SOURCE ASSESSMENT

Best Buy Co., Inc is a specialty electronics retailer in the United States of America. It started as a small business by the name Sound of Music and has now become one of the biggest and most popular electronics retailer in America and beyond. Best Buy offers a wide range of technology products, including computers, laptops, and home appliances or devices. These devices can be acquired online through the company's e-commerce platform or by visiting any of the company's numerous brick and mortar stores. Best Buy also provides technical services such as device and software installation, repair of devices, and device configuration, among others, for all their customers. This is done by the Geek Squad, a team of IT technicians who are responsible for all technical support services. Best Buy now has over 85,000 employees and over 1000 stores in America and Canada (Best Buy, 2025). The headquarters is located at Richfield, Minnesota, and the Chief Executive is currently Corrie Barry (Databahn, 2025).

Best Buy is a high tech company and , as such, relies heavily on technology including software in their operations. These operations include Customer Relations Management(CRM), information or cyber-security, inventory management, e-commerce optimization, data analytics and business intelligence. With the cost of operations on the rise, businesses like Best Buy are constantly in search of new ways or strategies to reduce expenses and optimize workflows; one of the most effective approaches they've embraced is the adoption of Open Source Software (OSS). This report includes a detailed analysis of Open Source Software, including a threat and vulnerability assessment as well as the pros and cons of acquiring and utilizing them.

Open source software (OSS) has been defined as "software that is distributed with its source code, making it available for use, modification, and distribution with its original rights" (Blackduck, 2025). As this definition suggests, OSS is shared publicly and anyone can have access to them, modify them, use and redistribute them provided they adhere to the license's terms. OSS, therefore presents a very feasible option for businesses like Best Buy. However, there are certain risks that pertain to it, which could have very damaging effects on Best Buy's profitability and survivability. It is therefore important that these threats are identified and mitigated against. These threats are rated on the potential impacts as well as their probability of occurrence using a tool known as the threat matrix. This matrix helps organizations and professionals prioritize risks by rating their probability and severity. A high rating on the matrix indicates a threat that is both likely to happen and capable of causing significant damage and hence needs to be addressed immediately and properly. On the other hand, a low rating indicates a threat that is less likely to happen and not capable of causing significant damage to the organization. Below is an example of the matrix.

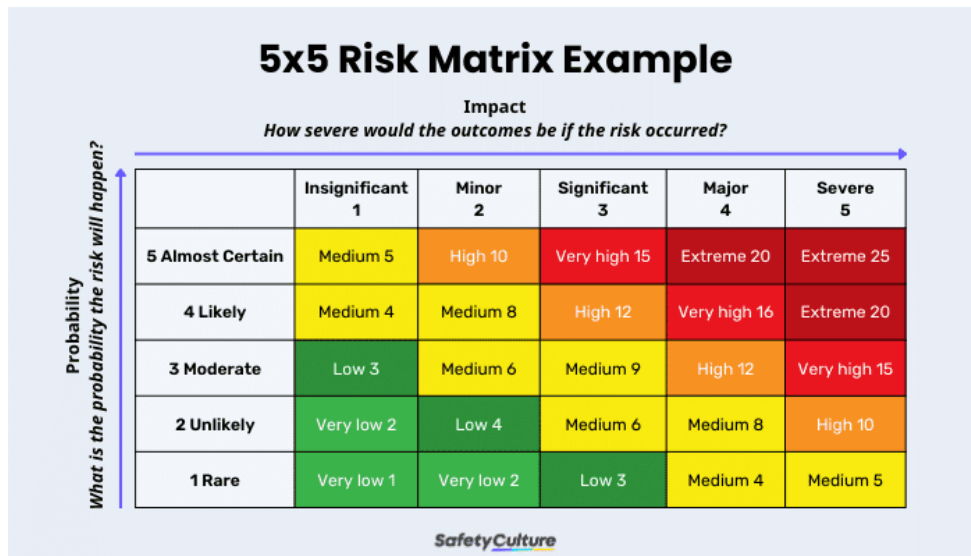


Image from Safety culture, 2025

One of the threats to Open Source Software is Malware. This has a rating of 20 on the matrix. this makes it a high scoring threat. “Malware, short for malicious software, refers to any intrusive software developed by cybercriminals (often called hackers) to steal data and damage or destroy computers and computer systems” (Cisco, 2025). Malware includes viruses. They are malicious programs that need a host to spread and they are able to modify systems as well as programs to include working copies of themselves. Another form of malware is worms, which are self-replicating and unlike viruses, do not need a host to propagate. Trojans are another form of malware. These forms of malware may appear innocuous but have concealed motives. Ransomware, another form of malware prevents legitimate access until an amount is paid. Malware also includes Botnet, a network of computer systems controlled by malware and used to perform attacks on computer systems. The source code for Open Source Software is accessible by the general public and this opens the door for an adversary to access and assess the source codes of these software packages for vulnerabilities and then launch an attack. Also an attacker could pose as a contributor and submit a malicious code in an existing project which might go unnoticed if not well examined. If Best Buy uses an OSS which has been compromised in this way, it could be a victim of a malware attack and the effects of could this impact the profitability and the survivability heavily.

Another threat to Open Source Software is hacking. This has a rating of 20. This makes it a high rating threat and as such, likely to happen with its consequence being very unwanted. Hacking has been defined as “ the unconventional use or illicit means to gain unauthorized access to a digital device, computer system or network” (IBM, 2024). Actors of this crime could gain access into organizations’ system and steal data as well as engage in ransomware. As mentioned earlier, the public access of the source code of OSS allows perpetrators of this crime to access the source code of programs, examine the code for loopholes and launch an attack. Also, the availability and accessibility of the source code allows these evil doers to compare existing programs with similar design whose vulnerabilities they already know and through that launch an attack. These attackers could steal customer data and cause e-commerce downtime that can damage the hard-earned reputation of brands like Best Buy.

Open Source Software could also enable a Denial of Service Attack. “A distributed denial-of-service (DDoS) attack is a malicious attempt to disrupt the normal traffic of a targeted server, service or network by overwhelming the target or its surrounding infrastructure with a flood of Internet traffic” (Cloudflare, 2025). DDoS essentially reduces or denies legitimate use of server or network resources. Twitter, Facebook, LiveJournal and some Google services were victims of a DDoS attack in 2009. This disrupted their servers and rendered them offline (The Guardian, 2009). As mentioned earlier, the source code and all of its components are publicly known. This means that all the vulnerabilities in the code could also be known and exploited by an adversary to launch a DDoS attack. Like Twitter, Facebook and LiveJournal, Best Buy could have services like the e-commerce platform disrupted if it falls victim to a DDoS attack which could result in loss of huge sums of money and also destroy its hard-earned reputation.

The aforementioned threats can be mitigated by ensuring that software is kept up-to-date at all times with the latest patches, installing and keeping firewalls updated, and educating users of computer systems and software on the best practices and why it is necessary to keep all software up-to-date. Also, a vulnerability scanner like Nessus could be used to scan for vulnerabilities in the systems and software packages.

Open Source Software has some very essential benefits. One of them is that it saves cost. Cost-effectiveness is one of the most attractive or appealing features of OSS to businesses including Best Buy. OSS requires no licensing fees associated with proprietary software, so there’s no need to pay an amount to acquire them and there’s no need to worry about vendors increasing the prices of software packages over time. OSS also presents no extra costs like upgrade fees, premium support charges and customization charges. The cost-effectiveness of OSS reduces the overall project costs. Best Buy could use savings from adopting OSS to enhance other areas of the business or support expansion.

Secondly, OSS enhances flexibility and customization. Unlike proprietary software, OSS allows for its source code to be accessed and tailored to meet the exact business needs of users. Best Buy can therefore modify an OSS solution to meet needs inherent to the company. OSS also allows for bug fixing and the addition of new features which makes them adaptable to future situations.

Thirdly, OSS ensures improved security and transparency. OSS makes the source code of software solutions available for vetting and scrutiny. This ensures that vulnerabilities and other issues are identified and rectified or patched. This transparency, a key fundamental of OSS, ensures that there is a fast response to security issues. According to Bairesdev, “because of this transparency, the velocity in which bugs and threats are addressed is under eight hours, compared to 6.9 days for developers working with proprietary software. Oh, and those almost seven days begin to count *only* after the bugs have been found and reported, since many of them are not encountered until the product has been released to the masses” (BairesDev, 2025).

Fourthly, OSS avoids vendor lock-ins which often characterize proprietary software packages. An organization may become overly reliant on the developer or vendor of a proprietary software for new features, bug fixes, updates, and technical support. OSS eliminates that dependency and gives users including Best Buy full control over its technology stack. This allows for the adaptation of new technologies and services or service providers.

Lastly, OSS attracts and retains talents. The flexibility and innovation that OSS like python, Apache, OpenProject and MySQL presents appeals more to data scientists, program developers and engineers. These professionals therefore lean towards businesses or companies that use open source solutions. As one of the leading retail brands in America, Best Buy could further expand its talent pool and strengthen its competitive edge against companies like Amazon and Walmart by adopting open source software solutions.

Despite these advantages, OSS faces some issues that organizations have to consider before adopting them. One of them is its security issues. As discussed earlier, OSS makes the source code available publicly and as such perpetrators of cyber-attacks could thoroughly access the source code for vulnerabilities to launch an attack. The Log4j vulnerability, also known as Log4Shell is an example of a vulnerability that was discovered in the logging library in November 2021. It essentially grants hackers total control of devices running an unpatched version of Log4j (IBM, 2025). For a major retailer like Best Buy, the effect of an attack stemming from the exploitation of a vulnerability in OSS this could have huge ramifications, affecting not only the profitability but also the long term survivability and the trustworthiness of the brand.

Secondly, OSS lacks dedicated support. OSS lacks the 24/7 technical support that proprietary software packages offer. Businesses and users in general may therefore have to rely on community-driven support channels. These channels may not be reliable and timely. This may affect Best Buy immensely as it operates in a very fast paced and competitive retail environment and every second counts. Every downtime may lead to loss of thousands or even millions of dollars.

Thirdly, OSS maintains some hidden costs. It is true that OSS provides some cost-effective or free software solutions to organization and users in general but it comes with some hidden or unexpected costs. Best Buy would have to spend an amount to employ people capable of working with OSS and also train existing staff to work with OSS. Additionally, Best Buy may have to incur extra cost to customize the software to suit its business needs. There may also be hidden cost related to security updates and third-party technical support services.

Fourthly, the licensing of OSS could be complex. Some of the license requirements for OSS could be very complex making it challenging for businesses like Best Buy to fully understand their obligations. Businesses or users therefore, need to carefully evaluate the terms and conditions of these licenses for OSS and ensure that they are complied with to avoid legal breaches. Also, some of the terms stated by the licenses may not align with Best Buy's policies and, as such, must be well scrutinized to avoid infringements.

Lastly, there could be issues with integration and compatibility. Some OSS solutions may not be compatible with existing infrastructure in Best Buy. This could mean that an extra amount may have to be invested in updating or configuring systems to work with OSS. Alternatively, even more amount may be spent on acquiring new infrastructure that are compatible with OSS solutions.

In conclusion, Open Source Software has some huge benefits including cost-savings, talent acquisition and flexibility that Best Buy, an institution that relies and heavily invests in software solutions, could benefit from in today's competitive retail market . As mentioned earlier, OSS also comes with some drawbacks which could affect Best Buy negatively. However, I believe most these drawbacks could be managed if the appropriate steps are taken. For example, to avoid security issues, all systems and software must be kept up to

date at all times so the probability and the impacts of these threats are reduced to the barest minimum. Furthermore, frequent code audits and employee training could also help minimize the probability of cyber-attacks. Also a license compliance team could be established to work with the IT department to ensure that all licenses are properly evaluated and complied with.

I therefore firmly believe that benefits of acquiring and utilizing Open Source Software outweighs the disadvantages and therefore recommend that Best Buy acquires and properly implements them so the most is made out of them.

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